HANDWRITTEN STUDY NOTES

GEOSYNTHETICS AND REINFORCED EARTH STRUCTURES

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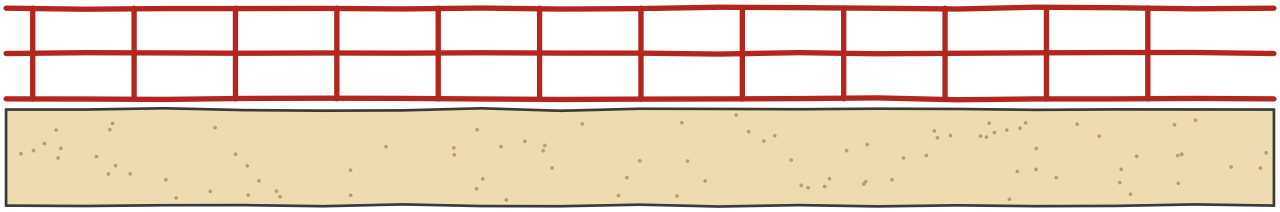
UNIT – IV

Reinforced Embankments

Syllabus topics covered in these notes:

- Reinforced embankments: introduction, design of reinforced embankment
- Foundation mattress below the embankment: purpose, function, components
- Design of reinforced mattress: settlement control, bearing capacity, long-term performance
- Field implementation and monitoring techniques

soil + reinforcement = strong composite



1. Reinforced Embankments – Introduction

Embankments for highways, railways, flood banks and approaches are often built on **soft, compressible clays** (low undrained strength c_u, high compressibility). An unreinforced embankment can fail by slip, spreading or bearing failure, or settle excessively. **Basal reinforcement** (high-strength geotextile / geogrid) laid at the base of the fill – the **foundation mattress** – raises stability and allows faster, higher construction.

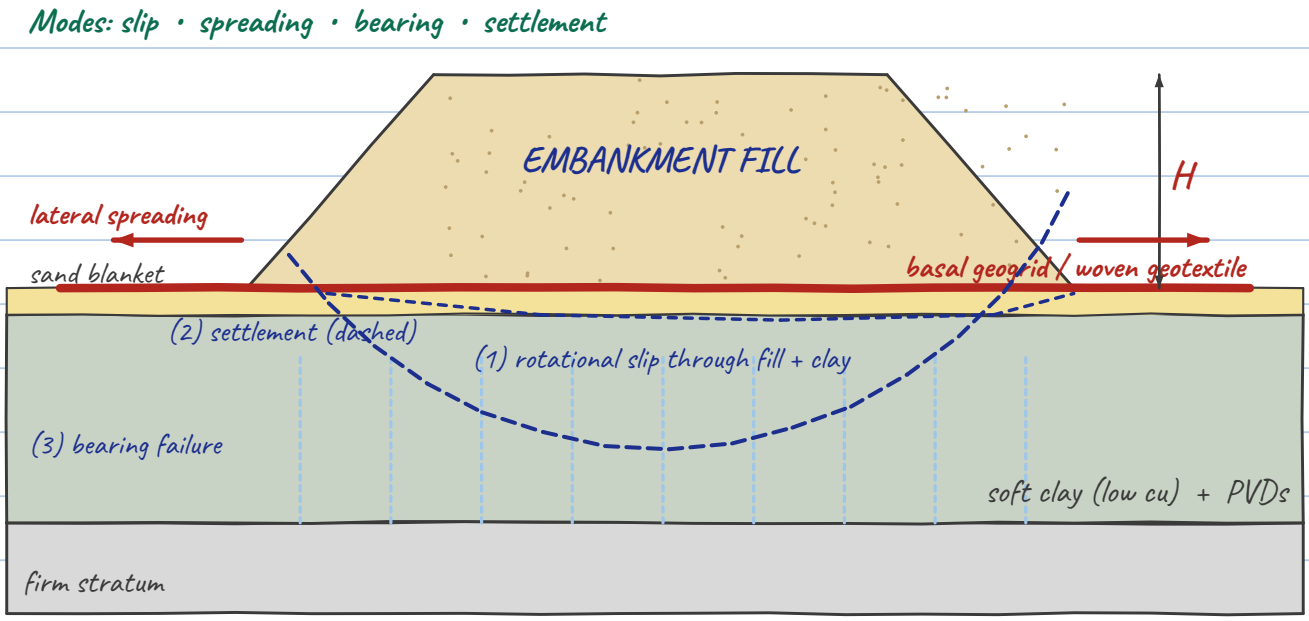


Fig 4.1 Embankment on soft ground with basal reinforcement

Failure modes to be checked

1. **Rotational (slope) failure** through fill and soft foundation – needs reinforcement tension T to resist moment.
2. **Lateral spreading / sliding** of the fill over the base – outward thrust $P_a = \frac{1}{2} K_a \gamma H^2$ resisted by tension and friction.
3. **Foundation (bearing) failure** – squeezing of soft clay: $q_{ult} \approx 5.14\, c_u$.
4. **Excessive total / differential settlement** and long-term consolidation.
5. **Reinforcement rupture, pull-out, or excessive strain** (> 5–6 %).

Types of reinforced embankment

- **Basal-reinforced embankment** on soft soil (geotextile / geogrid at the base).
- **Reinforced steep side-slope embankment** (reinforced soil slope, layers every 0.3–0.6 m to angles up to 70°) to save land.
- **Piled embankment** with geosynthetic platform bridging piles / stone columns.
- **Staged construction** with PVDs and preloading (reinforcement + drains).

2. Design of Reinforced Embankment

2.1 Data and requirements

- Embankment height H, side slope, fill properties (γ, φ), traffic surcharge q.
- Foundation soil: c_u profile, thickness of soft layer, C_c, e_0, c_v, c_h, water table.
- Required FS: slope stability ≥ 1.3 (short term) / 1.5 (long term); spreading ≥ 1.5; bearing ≥ 1.3–1.5.

2.2 Unreinforced critical height

$$H_c \approx 5.14\, c_u / \gamma$$

Example 1

c_u = 15 kPa, γ_{fill} = 19 kN/m³ → H_c = 5.14 × 15 / 19 = 4.06 m; safe height for FS = 1.3: ≈ 3.1 m.

If the embankment must be 6 m high → reinforcement / staging / drains are needed.

2.3 Stability against lateral spreading

Outward thrust of the fill (active) is resisted by the **tension in the basal geosynthetic**:

$$T_{req} \geq FS \cdot \tfrac{1}{2} K_a \gamma H^2, \quad K_a = \tan^2(45° - \varphi/2)$$

Also check **sliding of fill over reinforcement**: $FS = (\gamma H L \tan\delta) / (\tfrac{1}{2} K_a \gamma H^2) \geq 1.5$ and **pull-out / anchorage** of the geosynthetic beyond the failure zone.

Example 2

$H = 5$ m, $\gamma = 19$, $\varphi = 30°$ ($K_a = 0.333$): $P_a = ½ \times 0.333 \times 19 \times 25 = 79$ kN/m

$T_{req} = 1.5 \times 79 = 118$ kN/m (long-term allowable) → $T_{ult} = 118 \times (RF_{CR} \cdot RF_{ID} \cdot RF_D \approx 2.4) \approx 285$ kN/m → use PET geogrid / woven GT 300 kN/m.

2.4 Global (rotational) stability

- Use **modified Bishop** with undrained strength (total stress) for short term; reinforcement force T acts at its level and **adds a resisting moment $T \cdot y$**.
- Unreinforced FS_u from search of critical circle (e.g. 0.9). Required extra tension:

$$T_{req} = (FS_{req} - FS_u) \cdot M_D / R$$

M_D = driving moment, R = lever arm (≈ radius if reinforcement is horizontal). The most critical circles often pass through the soft clay below the mattress, so also check deep circles and the reinforcement anchorage.

2.5 Other design checks

- **Reinforcement strain** at end of construction limited to ≈ 5–6 % (≤ 2 % residual for long-term); stiffness $J \geq T/\varepsilon$.
- **Embankment fill**: granular ($\varphi \geq 30°$), compacted ≥ 95 % MDD; side slopes protected against erosion.
- **Anchorage** length beyond the toe: wrap-around (return 1–2 m) or trench anchor.

2.6 Stage construction and strength gain

When the fill is raised in stages with waiting periods, pore pressure dissipates and the clay gains strength, which allows a higher final embankment:

$$\Delta c_u = (c_u / \sigma'_v) \cdot U \cdot \Delta\sigma'_v$$

Example 5

$c_u/\sigma'_v = 0.25$, $\Delta\sigma = 60$ kPa, $U = 70$ % after the first stage → $\Delta c_u = 0.25 \times 0.70 \times 60 = 10.5$ kPa

New $c_u = 15 + 10.5 = 25.5$ kPa → $H_c = 5.14 \times 25.5 / 19 = 6.9$ m (up from 4.1 m).

2.7 Reinforced steep side slopes

- Layers of geogrid / geotextile (spacing 0.3–0.6 m) with wrap-around or facing give slopes up to ≈ 70°, saving land width.
- Design like a wall: $T = K\sigma_v S_v$; check rupture, pull-out beyond the critical slip surface (log-spiral / circular), and local stability between layers.
- Secondary short layers near the face improve compaction and erosion resistance; use drainage and vegetation / erosion mats on the face.

3. Foundation Mattress below the Embankment

3.1 Purpose and function

- **Reinforcement** – takes tension, prevents lateral spreading and increases slope stability.
- **Load distribution** – spreads embankment load (about 2V:1H) so stress on soft soil is lower and more uniform.
- **Reduces differential settlement** and bridges local weak spots / voids.
- **Bearing capacity improvement** – tensioned-membrane effect and confinement.
- **Separation** – prevents mixing of fill and clay; **working platform** for equipment on very soft soil.
- **Drainage** – granular layer (with PVDs) provides a horizontal drainage path so consolidation is faster.

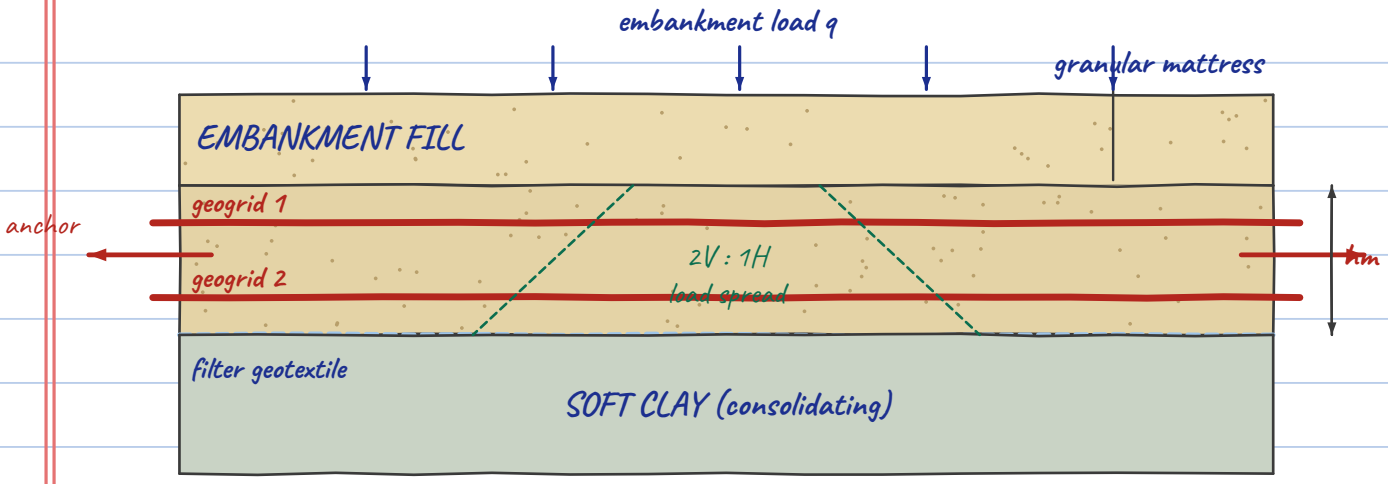


Fig 4.2 Reinforced granular foundation mattress

3.2 Components

1. **Geosynthetic reinforcement** – high-strength woven geotextile (PET / PP / PVA) or geogrid, 1–3 layers; strength 50–500 kN/m, low creep, strain < 5 %.
2. **Granular layer** – sand / gravel 0.3–1.0 m thick, $\varphi \geq 35^\circ$, compacted; also acts as drainage blanket.
3. **Separation / filter geotextile** (nonwoven) between soft clay and granular layer.
4. **Prefabricated vertical drains** (optional) connected to the blanket.
5. **Edge details** – wrap-around or anchor trench; toe berms; drainage outlets.

Variants: **geocell mattress** (3-D cells filled with aggregate, stiff slab-like), **fascine / brushwood** mattress (traditional), **piled mattress** over columns.

4. Design of Reinforced Mattress

Steps: (1) find T required for stability and spreading, (2) choose reinforcement and layers, (3) fix mattress thickness h_m, (4) check settlement, (5) bearing capacity, (6) long-term performance.

4.1 Mattress thickness and load spread

$$\Delta\sigma \text{ at clay top} = q\,B / (B + h_m) \quad \text{(2V:1H spread, strip load B)}$$

Typical h_m = 0.5–1.0 m; thicker layer → better load spreading but more weight; choose so that stress on soft clay stays below ≈ 5 c_u/FS.

4.2 Settlement control

Primary consolidation settlement of the soft clay (thickness H_0):

$$\rho_c = [C_c H_0 / (1 + e_0)] \cdot \log[(\sigma'_0 + \Delta\sigma') / \sigma'_0]$$

$$\rho_s = [C_\alpha H_0 / (1 + e_p)] \cdot \log(t_2/t_1) \text{ (secondary)}$$

Example 3

C_c = 0.45, e_0 = 1.2, H_0 = 6 m, σ'_0 = 40 kPa (mid-depth), $\Delta\sigma'$ = 60 kPa

ρ_c = (0.45 × 6 / 2.2) × log(100/40) = 1.227 × 0.398 = **0.49 m**

Time: $t = T_v H_{dr}^2 / c_v$; for U = 90 % (T_v = 0.848), c_v = 1.5 m^2/yr: single drainage (H_{dr} = 6) → 20 yr; double (3 m) → **5.1 yr**; with PVDs → months.

- Reinforcement **does not stop consolidation settlement**; it makes it **uniform**. Control by **PVDs, preloading / surcharge, stage construction and lightweight fill**.
- Allowed settlement: ≈ 100–300 mm total residual after pavement for highways; differential < 1:200 – 1:100.

4.3 Bearing capacity

$$q_{ult} = c_u N_c + \Delta q_R, \; N_c = 5.14, \; \Delta q_R \approx 2T \sin\theta / B$$

Δq_R = increase due to tensioned membrane / confinement (T = mobilised tension, θ = deflection angle at the edge of the loaded area, B = width). Use FS ≥ 1.3 (embankment, temporary) to 2.5 (permanent structures). Reinforcement mainly improves **stability and uniformity** of bearing rather than large strength increase.

Tensioned membrane over a soft spot / void of width B_s and sag d:

$$T = q_m B_s^2 / (8d), \; \varepsilon \approx (8/3)(d/B_s)^2$$

Example 4

q_m = 10 kPa, B_s = 2 m, d = 0.25 m → T = 10 × 4 / (8 × 0.25) = **20 kN/m**; ε = 2.67 × $(0.125)^2$ = **4.2 %** (< 5 % OK).

4.4 Long-term performance

- **Creep:** $T_{al} = T_{ult} / (RF_{CR} \cdot RF_{ID} \cdot RF_{D})$; PET preferred for sustained load; limit long-term strain to ≈ 2–3 %.
- **Durability:** pH, temperature, chemicals; hydrolysis (PET in alkaline soils), UV during construction.
- **Consolidation and secondary compression** – raise embankment (over-build) to compensate; add maintenance allowance.
- **Drainage clogging** – filter geotextile compatibility; blanket outlets kept open.
- **Strength gain** of clay with consolidation improves FS with time (stage construction exploits this).

4.5 Geocell and piled alternatives

- **Geocell mattress:** honeycomb cells (height 150–300 mm) filled with granular soil act like a stiff **slab**; confinement increases friction angle and spreads load widely – excellent over very soft clay (c_u < 10 kPa); thickness of mattress ≈ 0.3–0.6 m.
- **Piled embankment:** pile caps / stone columns with 1–3 layers of geogrid forming a **load-transfer platform**; arching in fill transfers load to piles; used where settlement must be almost nil (bridge approaches).
- Choice: basal geotextile + PVD (cheap, time available) → geocell (very soft, low fill) → piles (zero settlement, little time).

Quick recap

Design summary for a mattress: (1) T_{req} from spreading & circle → (2) choose $T_{al} \geq T_{req}$ (after reduction factors) → (3) h_m = 0.5–1 m granular → (4) check $q_{ult}/q \geq 1.3$ → (5) predict ρ and time (PVDs) → (6) specify monitoring and limits.

5. Field Implementation

1. **Site preparation:** clear vegetation, cut trees at ground level, **leave the root mat**; fill holes; avoid disturbing very soft soil (use light equipment / mats).

2. Place **nonwoven separator** and **sand blanket**; install **PVDs** and connect to the blanket (if used).
3. **Unroll geosynthetic** perpendicular to the embankment axis (main strength direction across the embankment), pull taut, remove wrinkles; **overlap 0.3–1 m** or sew joints (CBR < 1 → sew / 1 m).
4. **Anchor** with pins / sandbags; place **fill from the edges toward the centre** (end-tipping over a stable sub-layer) so a mud-wave is not formed; first lift ≥ **0.3–0.5 m**, use low-ground-pressure dozers; **no direct traffic on bare geosynthetic**.
5. **Compact** only after first lift thickness is placed; later lifts 200–300 mm to ≥ 95 % MDD; wrap-around at slopes; **stage the rate of filling** (e.g. 0.3–0.5 m/week) guided by monitoring.
6. Complete drainage, erosion protection of slopes and settle-out period before pavement.

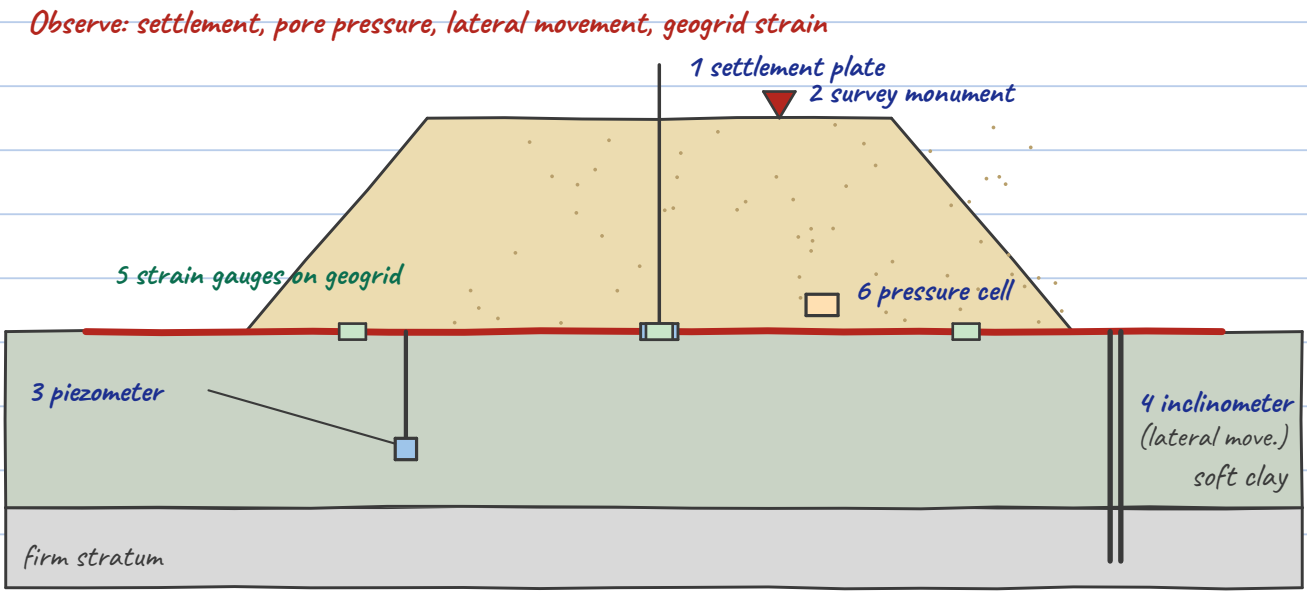


Fig 4.3 Instrumentation of a reinforced embankment

6. Monitoring Techniques

The **observational method** is used on soft soils: measure, compare with design predictions, and adjust the rate of filling.

Instrument	Measures	Purpose
Settlement plate / gauge	Vertical settlement at base	Degree of consolidation, Asaoka
Survey monuments (surface)	Heave, settlement at crest / toe	Differential movement
Piezometer (vibrating wire)	Excess pore pressure Δu	Rate of fill, strength gain
Inclinometer	Lateral displacement at toe	Stability, spreading
Strain gauge / tell-tale	Strain in geosynthetic	Tension T, creep, safety
Earth pressure cell	Stress at base	Load distribution
Extensometer	Differential vertical strain	Layer compression

Control criteria and prediction

- **Pore pressure ratio** $r_u = \Delta u / \Delta\sigma$ kept below ≈ 0.6–0.7 during filling; stop filling if it rises quickly.
- **Lateral-to-vertical movement** (Tavenas, Matsuo–Kawamura plots): a sudden increase of horizontal movement shows imminent failure; typical alert: $d\delta_h/dt >$ 2–3 mm/day.
- **Settlement prediction:** Asaoka method (plot ρ_i vs ρ_{i-1} at equal Δt; intersection with 45° line gives final settlement), hyperbolic method.
- **Geosynthetic strain** should remain < 5 %; creep strain should flatten with time.
- Frequency: daily during filling, weekly after, then monthly until settlement rate < 2–5 mm/month.

Advantages and limitations of reinforced embankments

- Allow construction on very soft soil (c_u < 15 kPa) without removal / replacement; **higher and steeper** embankments; faster construction and less fill.
- More uniform settlement, reduced cracking of pavement, less land take, lower cost than piling or soil replacement.

- Limitations: does not remove consolidation settlement; needs good quality control and monitoring; reinforcement cannot help if the failure surface is deep and passes below it.

Construction problems and remedies

Problem	Cause	Remedy
Mud-wave ahead of fill	Dumping on weak soil	Place from edges, thicker first lift
Wrinkles / slack	Poor tensioning	Pull taut, pin, sew joints
Geosynthetic damage	Direct traffic, sharp stones	Cover 0.3 m, select fill
High pore pressure	Fast filling	Stage filling, PVDs, wait
Edge failure / spreading	Weak anchorage	Wrap-around, longer anchorage

Case idea: embankment 6–8 m high on 8 m soft clay with PVDs at 1.5 m, two layers of 200 kN/m PET geotextile and a 0.6 m sand blanket; filled in two stages over 6 months and monitored by settlement plates and piezometers – a typical highway-approach solution.

Formula sheet – Unit IV

Item	Expression
Critical height	$H_c = 5.14\, c_u / \gamma$
Spreading tension	$T \geq FS \cdot \tfrac{1}{2} K_a \gamma H^2$
Extra tension (circle)	$T = (FS_{req} - FS_u)\, M_D / R$
Consolidation settlement	$\rho = C_c H_0 / (1+e_0) \cdot \log(\sigma'_1/\sigma'_0)$
Time	$t = T_v H_{dr}^2 / c_v$
Membrane tension	$T = q_m B_s^2 / (8\, d)$

Important questions

1. Explain the modes of failure of embankments on soft soil; how does reinforcement help?

2. What is a foundation mattress? Explain its purpose, function and components.
3. Describe the design of a reinforced mattress for settlement control and bearing capacity.
4. Discuss the long-term performance of reinforced embankments.
5. Explain field implementation and monitoring techniques with a neat sketch.